Algebra II with Trigonometry Chapter 5.4

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file: 4401270_alg-2trig-h-chp-5-4-note-docx.pdf

Solutions

Problems

1. Evaluate exactly:

$$\sin\left(2\arctan\frac{3}{4}\right)+\cos\left(2\arcsin\frac{3}{5}\right).$$

Let

$$\alpha=\arctan\frac{3}{4},\qquad \beta=\arcsin\frac{3}{5}.$$

Then for α, we may take a right triangle with legs $3, 4$, so hypotenuse 5. Hence

$$\sin\alpha=\frac{3}{5},\qquad \cos\alpha=\frac{4}{5}.$$

Therefore

$$\sin(2\alpha)=2\sin\alpha\cos\alpha=2\cdot\frac{3}{5}\cdot\frac{4}{5}=\frac{24}{25}.$$

For β, since $\sin\beta=\frac{3}{5}$, we have

$$\cos\beta=\frac{4}{5}$$

because $\beta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Thus

$$\cos(2\beta) = 1 - 2\sin^2\beta = 1 - 2 \cdot \frac{9}{25} = \frac{7}{25}.$$

So the required value is

$$\frac{24}{25} + \frac{7}{25} = \frac{31}{25}.$$

$$\boxed{\frac{31}{25}}$$

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2. Find the exact value of

$$\arctan\frac{1}{2} + \arctan\frac{1}{3} + \arctan\frac{1}{7}.$$

Let

$$A = \arctan\frac{1}{2}, \qquad B = \arctan\frac{1}{3}, \qquad C = \arctan\frac{1}{7}.$$

First combine A and B:

$$\tan(A+B) = \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} = \frac{5/6}{5/6} = 1.$$

Since $A, B > 0$ and $A + B < \frac{\pi}{2}$, it follows that

$$A + B = \arctan 1 = \frac{\pi}{4}.$$

Now add C:

$$\tan\left(\frac{\pi}{4} + C\right) = \frac{1 + \frac{1}{7}}{1 - 1 \cdot \frac{1}{7}} = \frac{8/7}{6/7} = \frac{4}{3}.$$

Hence

$$\frac{\pi}{4} + C = \arctan\frac{4}{3},$$

but there is an even simpler observation: if we compare with a 3-4-5 triangle, then

$$\arctan\frac{4}{3} = \arctan\left(\tan\theta\right)$$

for the acute angle θ with tangent $4/3$. So the sum equals that acute angle. To identify it more neatly, instead compute directly:

$$\tan(A+B+C) = \frac{1+\frac{1}{7}}{1-\frac{1}{7}} = \frac{4}{3}.$$

Since $A+B+C \in (0, \frac{\pi}{2})$, we conclude

$$A+B+C = \arctan\frac{4}{3}.$$

Thus

$$\boxed{\arctan\frac{4}{3}}.$$

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3. Let

$$\theta = \arcsin\frac{5}{13} + \arccos\frac{12}{13}.$$

Find $\tan\theta$ **exactly.**

Set

$$\alpha = \arcsin\frac{5}{13}, \qquad \beta = \arccos\frac{12}{13}.$$

Then

$$\sin\alpha = \frac{5}{13}, \quad \cos\alpha = \frac{12}{13},$$

since $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Also

$$\cos\beta = \frac{12}{13}, \quad \sin\beta = \frac{5}{13},$$

since $\beta \in [0, \pi]$ and here β is acute.

Thus $\alpha = \beta$, so

$$\theta = 2\alpha.$$

Therefore

$$\tan\theta = \tan(2\alpha) = \frac{2\tan\alpha}{1 - \tan^2\alpha}.$$

Now

$$\tan\alpha = \frac{5}{12},$$

so

$$\tan\theta = \frac{2 \cdot \frac{5}{12}}{1 - \left(\frac{5}{12}\right)^2} = \frac{10/12}{1 - 25/144} = \frac{5/6}{119/144} = \frac{5}{6} \cdot \frac{144}{119} = \frac{120}{119}.$$

Hence

$$\boxed{\frac{120}{119}}.$$

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4. A pole stands vertically on level ground. From point A, the angle of elevation to its top is 45°. After walking 20 meters directly away from the pole to point B, the angle of elevation becomes $\arctan\frac{2}{3}$. Find the height of the pole.

Let the height of the pole be h meters, and let the horizontal distance from A to the pole be x meters.

From point A,

$$\tan 45^\circ = \frac{h}{x} = 1,$$

so

$$h = x.$$

After walking 20 meters away, the distance from B to the pole is $x + 20$. From point B,

$$\tan\left(\arctan\frac{2}{3}\right) = \frac{h}{x+20} = \frac{2}{3}.$$

Thus

$$h = \frac{2}{3}(x+20).$$

Using $h = x$, we get

$$x = \frac{2}{3}(x+20).$$

So

$$3x = 2x + 40,$$

hence

$$x = 40.$$

Therefore

$$h = x = 40.$$

$$\boxed{40 \text{ meters}}$$

5. For $x \in [-1,1]$, simplify exactly:

$$\sin\bigl(2\arccos x\bigr) + \cos\bigl(2\arcsin x\bigr).$$

Let

$$\alpha = \arccos x, \qquad \beta = \arcsin x.$$

Since $\alpha \in [0,\pi]$, we have

$$\sin\alpha = \sqrt{1-x^2},$$

because $\sin\alpha \geq 0$. Hence

$$\sin(2\alpha) = 2\sin\alpha\cos\alpha = 2x\sqrt{1-x^2}.$$

Also

$$\cos(2\beta) = 1 - 2\sin^2\beta = 1 - 2x^2.$$

Therefore the expression simplifies to

$$2x\sqrt{1-x^2} + 1 - 2x^2.$$

So the exact simplification is

$$\boxed{1 - 2x^2 + 2x\sqrt{1-x^2}}.$$

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6. Find all $\theta \in [0,\pi]$ satisfying

$$\arcsin(\sin\theta) = \arccos(\cos\theta).$$

We use the principal-value ranges:

$$\arcsin(\sin\theta) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \qquad \arccos(\cos\theta) \in [0,\pi].$$

Now for $\theta \in [0,\pi]$, we have

$$\arccos(\cos\theta) = \theta,$$

because $\arccos$ is the inverse of $\cos$ on $[0, \pi]$.

So the equation becomes

$$\arcsin(\sin\theta) = \theta.$$

But $\arcsin(\sin\theta) = \theta$ exactly when $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Intersecting with $[0, \pi]$, we get

$$\theta \in \left[0, \frac{\pi}{2}\right].$$

Indeed, for $\theta \in [0, \frac{\pi}{2}]$, both sides equal θ. For $\theta \in (\frac{\pi}{2}, \pi]$,

$$\arcsin(\sin\theta) = \pi - \theta \neq \theta$$

except possibly at $\theta = \frac{\pi}{2}$, already included.

Thus all solutions are

$$\boxed{\theta \in \left[0, \frac{\pi}{2}\right]}.$$